Top 50 Azure Fundamentals Questions

with Real-World Scenarios & Answers

1. What is Microsoft Azure ?

Scenario : Your company wants to host applications without purchasing physical servers.

Answer : Azure is Microsoft's cloud computing platform that provides on-demand compute, storage, networking, databases, analytics and AI services.

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2. What are the three main cloud service models ?

Scenario : You need a virtual machine, a managed platform or a ready-to-use email service.

Answer : IaaS provides infrastructure, PaaS provides a managed development platform and SaaS delivers ready-to-use software.

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3. What is the difference between Public, Private and Hybrid Cloud ?

Scenario : A bank keeps sensitive workloads on-premises but runs other applications in Azure.

Answer : Public cloud uses a provider's shared infrastructure, private cloud is dedicated to an organization and hybrid cloud combines on-premises or private infrastructure with public cloud.

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4. What is an Azure Region ?

Scenario : Your Indian customers need low-latency access to an application.

Answer : An Azure Region is a geogfraphic area containing one or more datacenters where Azure services are deployed.

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5. What is an Availability Zone ?

Scenario : An application must continue running if one datacenter experiences an outage.

Answer : Availability Zones are physically separate locations within a supported Azure region, designed to improve resilience against datacenter-level failures.

Key Terms to Remember :

1. **Region** → Geographic area with one or more datacenters.
2. **Availability Zone** → Physically separate location within a region.
3. **Subscription** → Management and billing boundary.
4. **Resource Group** → Logical container for related resources.

Cloud Service Models :

- **IaaS** → Infrastructure (eg. Virtual Machines)
- **PaaS** → Managed platform (eg. App Service)
- **SaaS** → Ready-to-use software (eg. Outlook, Teams)

6. What is an Azure Subscription ?

Scenario : Your organization wants to separate development and production resource usage and billing.

Answer : An Azure Subscription is a management and billing boundary for Azure resources, governed by applicable access and policy controls.

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7. What is a Resource Group ?

Scenario : Your data pipeline contains a storage account, Data Factory and monitoring resources.

Answer : A Resource Group is a logical container for managing related Azure resources together.

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8. What is Azure Resource Manager (ARM) ?

Scenario : You need to deploy multiple Azure resources consistently using templates.

Answer : Azure Resource Manager (ARM) is Azure's deployment and management layer for creating, updating, organizing and controlling resources.

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9. What is the difference between Scalability and Elasticity ?

Scenario : An application needs more capacity during peak traffic and less at night.

Answer : Scalability increases or decreases capacity, while elasticity automatically adjusts capacity in response to changing demand.

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10. What is the Shared Responsibility Model ?

Scenario : Your company migrates a database to a managed Azure service.

Answer : Microsoft secures the cloud infrastructure, while customers remain responsible for their assigned security duties, which vary by service model.

11. What is an Azure Virtual Machine (VM)?

Scenario : You need full control over the operating system to install custom software for your application.

Answer : An Azure Virtual Machine (VM) is an on-demand virtual server that provides full control over the operating system, installed software and configuration.

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12. What is Azure App Service ?

Scenario : You want to host a web application or REST API without managing virtual machines.

Answer : Azure App Service is a fully managed platform for hosting web apps, REST APIs and mobile backends.

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13. What is Azure Functions ?

Scenario : A file is uploaded to Blob Storage and you want to run a small piece of code to process the file.

Answer : Azure Functions is a serverless, event-driven compute service that runs code without managing servers.

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14. What is Azure Virtual Machine Scale Sets?

Scenario : Your application needs multiple VM instances during high traffic and should automatically scale.

Answer : Azure Virtual Machine Scale Sets automatically deploy, manage and scale a group of identical VMs based on demand.

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15. What is Azure Kubernetes Service (AKS) ?

Scenario : Your organization runs containerized microservices and needs a managed Kubernetes platform.

Answer : Azure Kubernetes Service (AKS) is a managed Kubernetes service for deploying, scaling and managing containerized applications.

16. What is Azure Container Instances (ACI)?

Scenario : You want to run a container quickly without managing virtual machines or a Kubernetes cluster.

Answer : Azure Container Instances (ACI) runs containers directly in Azure without requiring you to manage virtual machines or a Kubernetes control plane.

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17. What is Azure Storage Account ?

Scenario : Your application needs a centralized account to store different types of data like files, logs and backups.

Answer : A Storage Account provides a namespace and configuration boundary for Azure Storage services such as Blob, File, Queue and Table storage.

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18. What is Azure Blob Storage ?

Scenario : You need to store unstructured data such as images, videos, logs or backup files.

Answer : Azure Blob Storage is an object storage service designed to store large amounts of unstructured data.

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19. What is Azure Data Lake Storage Gen2 (ADLS Gen2)?

Scenario : Your data engineering team stores raw, cleansed and curated datasets for analytics and big data processing.

Answer : Azure Data Lake Storage Gen2 is a high-performance storage service derived from Blob Storage, with hierarchical namespace and data-lake-optimized features for analytics.

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20. What is the difference between Blob Storage and ADLS Gen2 ?

Scenario : You need to choose the right storage for general file storage vs. a data lake for analytics.

Answer : Blob Storage is for general-purpose object storage, while ADLS Gen2 provides hierarchical namespace, better access controls and data-lake capabilities for big data analytics.

21. What are Azure Blob Storage Access Tiers ?

Scenario : You need to store frequently accessed data and also long-term backup data with cost optimization.

Answer : Azure Blob Storage provides four access tiers - Hot, Cool, Cold and Archive, each offering different storage and access cost trade-offs based on how frequently the data is accessed.

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22. What is Azure Files ?

Scenario : Multiple virtual machines need a shared file storage that can be accessed using SMB or NFS protocol.

Answer : Azure Files provides managed file shares in the cloud that can be accessed via standard file sharing protocols like SMB and NFS.

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23. What is Azure Queue Storage ?

Scenario : Your application needs to send background tasks to be processed asynchronously.

Answer : Azure Queue Storage is a message queue service that allows you to store messages for asynchronous processing between application components.

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24. What is Azure Table Storage ?

Scenario : You need a NoSQL data store to store large amounts of structured, non-relational data for an application.

Answer : Azure Table Storage is a NoSQL key-value store for storing large volumes of structured data with a schema-less design.

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25. What are Azure Storage Redundancy Options ?

Scenario : Your business wants to ensure storage data remains available even if there is a hardware or regional failure.

Answer : Azure Storage provides LRS (Locally Redundant), ZRS (Zone Redundant), GRS (Geo-Redundant) and GZRS (Geo-Zone Redundant) options for different levels of durability and availability.

26. What is an Azure Virtual Network (VNet) ?

Scenario : Your application resources need a secure and isolated network in the cloud.

Answer : An Azure Virtual Network (VNet) is a logically isolated network in Azure that lets you securely connect Azure resources with each other and with on-premises networks.

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27. What is a Subnet ?

Scenario : You want to separate web servers and database servers within the same VNet.

Answer : A subnet is a range of IP addresses within a VNet that divides the network into smaller segments for better management and security control.

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28. What is a Network Security Group (NSG) ?

Scenario : You want to allow only specific traffic (like HTTP/HTTPS) and block all other inbound traffic to a VM.

Answer : A Network Security Group (NSG) contains security rules that allow or deny inbound and outbound network traffic to Azure resources.

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29. What is Azure VPN Gateway ?

Scenario : Your on-premises office needs a secure connection to Azure over the internet.

Answer : Azure VPN Gateway provides encrypted VPN connections between your on-premises network and Azure over the public internet.

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30. What is Azure ExpressRoute ?

Scenario : Your organization needs a private, high-performance and reliable connection to Azure without using the public internet.

Answer : Azure ExpressRoute provides a private and dedicated network connection from your on-premises infrastructure to Azure through a connectivity provider.

31. What is Azure Load Balancer ?

Scenario : Your application has multiple virtual machines and you want to distribute incoming traffic across them for high availability.

Answer : Azure Load Balancer distributes inbound and outbound network traffic across multiple virtual machines or services to ensure high availability and scalability.

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32. What is Azure Application Gateway ?

Scenario : You want to manage web traffic for your application with features like SSL offloading, URL-based routing and a Web Application Firewall (WAF).

Answer : Azure Application Gateway is a web traffic load balancer that provides URL-based routing, SSL termination and WAF for secure and scalable web applications.

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33. What is Azure Private Endpoint ?

Scenario : Your application in a virtual network needs to securely access an Azure Storage account without using the public internet.

Answer : Azure Private Endpoint provides a private IP address in your VNet to connect securely to Azure services over the Microsoft backbone network.

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34. What is Microsoft Entra ID ?

Scenario : Your employees need to sign in to Azure resources and enterprise applications using a single identity.

Answer : Microsoft Entra ID (formerly Azure AD) is a cloud-based identity and access management service for managing users, applications and secure access.

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35. What is Azure Role-Based Access Control (RBAC) ?

Scenario : You want to give a data engineer read access to a storage account but prevent them from deleting resources.

Answer : Azure RBAC allows you to assign roles with specific permissions to users, groups or service principals at different scopes (subscription, resource group or resource)

36. What is Azure Key Vault ?

Scenario : Your application needs to store secrets, API keys, database credentials and certificates securely.

Answer : Azure Key Vault is a cloud service to securely store and manage secrets, keys, and certificates with fine-grained access controls.

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37. What is a Managed Identity ?

Scenario : An Azure Function or VM needs to access Key Vault or Storage without storing credentials in the code.

Answer : A Managed Identity provides an automatically managed identity for Azure resources to authenticate to other Azure services without managing credentials.

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38. What is Azure Policy ?

Scenario : Your organization wants to restrict the creation of resources to specific regions and enforce naming standards.

Answer : Azure Policy is a governance service that helps enforce organizational rules, audit compliance and apply allowed or denied configurations to Azure resources.

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39. What is Microsoft Defender for Cloud ?

Scenario : Your organization wants to identify security vulnerabilities and improve the overall security posture of Azure resources.

Answer : Microsoft Defender for Cloud provides security monitoring, threat detection, vulnerability assessment and recommendations for Azure resources.

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40. What is Multi-Factor Authentication (MFA) ?

Scenario : A user's password is compromised and you want to prevent unauthorized access to Azure resources.

Answer : Multi-Factor Authentication (MFA) requires an additional verification method (such as a phone, authenticator app or security key) along with the password to sign in.

41. What is Azure Monitor?

Scenario : Your application is running in Azure and you want to monitor its performance, set up alerts and analyze logs.

Answer : Azure Monitor is a comprehensive monitoring service that collects, analyzes and acts on telemetry data from Azure resources. It helps you monitor performance, set alerts, visualize metrics and troubleshoot issues across your Azure environment.

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42. What is Azure Cost Management?

Scenario : Your organization wants to track cloud spending, set budgets and identify cost optimization opportunities.

Answer : Azure Cost Management is a service that helps you monitor, analyze and optimize your Azure spending. It provides cost visibility, budgeting, alerts and recommendations to reduce cloud costs.

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43. What is Azure Resource Group?

Scenario : You need to organize and manage multiple related Azure resources for your project.

Answer : A Resource Group is a logical container in Azure that holds related resources for an application. It allows you to manage, deploy and delete resources together as a unit.

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44. What is Azure ARM Template?

Scenario : You want to deploy a set of Azure resources in a consistent and repeatable way using Infrastructure as Code (IaC).

Answer : An Azure Resource Manager (ARM) Template is a JSON file that defines the infrastructure and configuration for Azure resources. It enables automated and repeatable deployments using declarative syntax.

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45. What is Azure Automation?

Scenario : You want to automate routine operations like starting/stopping VMs, applying patches or running scripts.

Answer : Azure Automation is a service that helps you automate repetitive management tasks, configuration management, software updates and process automation across your

46. What is Azure Backup ?

Scenario : Your organization wants to protect important data and VMs from accidental deletion or disasters.

Answer : Azure Backup is a cloud-based service that provides secure and reliable backup for Azure VMs, databases, files and other workloads, with options for automated backups, retention policies and easy recovery.

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47. What is Azure Site Recovery (ASR) ?

Scenario : Your application must stay available even if the primary region goes down due to a disaster.

Answer : Azure Site Recovery (ASR) replicates your on-premises or Azure workload to a secondary region, enabling disaster recovery and business continuity with minimal downtime.

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48. What is Azure Advisor ?

Scenario : You want recommendations to improve security, performarce, reliability and cost efficiency for your Azure resources.

Answer : Azure Advisor is a personalized cloud consultant that analyzes your resources and provides actionable recommendations to optimize cost, improve security, enhance performance and increase reliability.

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49. What is Azure Service Health ?

Scenario : Your critical application is not working and you want to know if there is a service outage or planned maintenance from Microsoft.

Answer : Azure Service Health provides real-time information about the health, outages and planned maintenance of Azure services in the regions you use, helping you respond quickly to issues.

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50. What is Azure Arc ?

Scenario : Your organization wants to manage on-premises servers, multi-cloud resources and edge devices using Azure.

Answer : Azure Arc is a management solution that extends Azure management, security and governance to on-premises, multi-cloud and edge environments, allowing you to manage all your resources from a single Azure portal.